

Shri Shamrao Patil (Yadravkar) Educational & Charitable Trust's



Sharad Institute of Technology
College of Engineering, Yadrav (Ichalkaranji)

An Autonomous Institute

Department of Electronics and Computer Engineering

A
Mini Project Report
on
LASER SECURITY ALARM SYSTEM

SUBMITTED BY

Roll No.

Student Name

13	Miss Pooja Saraswat
11	Miss Srushti Amar Patil
25	Miss Namrata Wandre
29	Miss Sanika Tribhuvane

Under the Guidance of
Dr. P. S. PATIL

Department of Electronics and Computer Engineering,
Sharad Institute of Technology College of Engineering, Yadrav-Ichalkaranji

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Sharad Institute of Technology College of Engineering
Yadrav (Ichalkaranji) - 416121
An Autonomous Institute

(Approved by AICTE, New Delhi, Recognized by Government of Maharashtra & Affiliated to Dr.

Babasaheb Ambedkar Technological University)

DEPARTMENT OF ELECTRONICS AND COMPUTER ENGINEERING

This is to certify that,

13	Miss Pooja Saraswat
11	Miss Srushti Amar Patil
25	Miss Namrata Wandre
29	Miss Sanika Tribhuvane

Studying in SY Electronics and Computer has successfully completed the Mini Project entitled **“LASER SECURITY ALARM SYSTEM”** under the guidance and supervision of Mr. P. S. Patil during the academic year 2024-2025 Semester IV. This is a part of partial fulfillment of the requirement for award of degree of the Bachelor of Technology in Electronics and Computer Engineering.

Dr. P. S. Patil

Guide

Dept. of Electronics and Computer Engg.

SITCOE

Dr. P. S. Patil

Associate Professor and Head

Dept. of Electronics and Computer Engg.

SITCOE

Dr. S. A. Khot

Principal

SITCOE

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ABSTRACT

This project deals with a model of laser security alarm system design. Laser security systems used to be difficult to install and rarely available to anyone other than the super-rich. Now, there are dozens of different security systems on the market that utilize lasers and can effectively protect everything from small apartments and businesses to large areas of property. Most home laser security systems consist of two parts: a basic alarm unit and an infrared motion detector. Laser based security system is a type of security and alarm system that uses laser light and a light sensor. Why a laser to be used? It is known that a laser light goes through long distance without any scattering effect (disturbing) and it is only visible at source and the destination point so it can be used as mediator between source and destination but to analyse the source a sensor is need, here the use of LDR is applicable. Just analysis is not enough alerting should be done in general alerting is sound effect so here buzzer act as alerting. Making use of this, a laser security system is designed. Its working: There is a laser diode that generates the laser beam which continuously strikes over the Light dependent resister sensors. When any person crosses the path, it inhibits laser to reach LDR and the sensor generate a low which is read by controller to power on the buzzer.

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INTRODUCTION

Need of security is the basic necessity of any individual. The feeling that we are safe and everything around us is all right is imperative for a peaceful living. But in this unsafe world, when crime, terror and threats are on their peak, how can one attain that sense of security? Here, laser security system provides us with a solution and for this reason more and more people are installing them in order to stay safe and secure. Various electronic security systems can be used at home and other important working places for security and safety purposes. Laser Security alarm is a device used for security purposes. It has a wide application in fields of security and defense starting from the security of simple house hold material to a very high valued material of an organization. They once used to be expensive solutions for security needs. Owing to cost cutting and fast technological advancements, this form of security system is becoming more affordable. Lasers differ from other light sources in a few significant ways. There are two features that are important for security systems. Unlike a light bulb or flashlight, laser light doesn't spread out, it is a narrow beam. And laser light is essentially a single colour. Because laser light doesn't spread much, it can be sent it a long way and still have enough energy in a small area to trigger the security system detector. Because it's a single wavelength, it can put a blocking filter on the detector to let laser light through without letting background light onto the detector. Laser light travels in a straight line. For instance, to protect the front of the yard, putting the laser at one corner and the detector at the other corner would do the job. That's not a very practical configuration, though. More typically, if it is needed to protect the perimeter of a room, or at least the enhances. So laser security systems start with a laser pointing to a small mirror. The first mirror is angled to direct the beam to a second small mirror, and so on until the final mirror directs the beam to the detector. If the beam is interrupted anywhere between the laser and the detector, the electronics will put the warning signal.

LITERATURE SURVEY

[1] Laser based security system using Arduino UNO by Paramitha Mondal, Madhusree Mondal. The proposed system contains sensor, Arduino UNO, ESP Wi-Fi module, buzzer, LDR module. In this system once security system detects the intruder buzzer starts buzzing very loudly, spy camera takes the photo of the intruder and sends it to the registered email address for valid proof.

[2] Laser based security system for home by Harshal Hemane, Debarati Sen. In this proposed system mirrors are used to reflect the laser rays to cover the region surrounding home in all direction. When laser light is incident on a mirror, it gets reflected from one mirror to another and falls on LDR. If any of the laser ray is blocked the buzzer produces beep sound.

[3] Laser security alarm system by A.B.N.V. Prasad, K. Ravi Raj, K. Siva Ganesh, M. Lithin Siva swamy Naidu, N. Phaneendra. In this project based on the voltage drop across the LDR is considered to turn On and Off the transistor. If the voltage drop across the LDR is low transistor is turned Off, if the voltage across the LDR is high transistor is said to be turned On. Once the transistor is turned on buzzer produces beep sound.

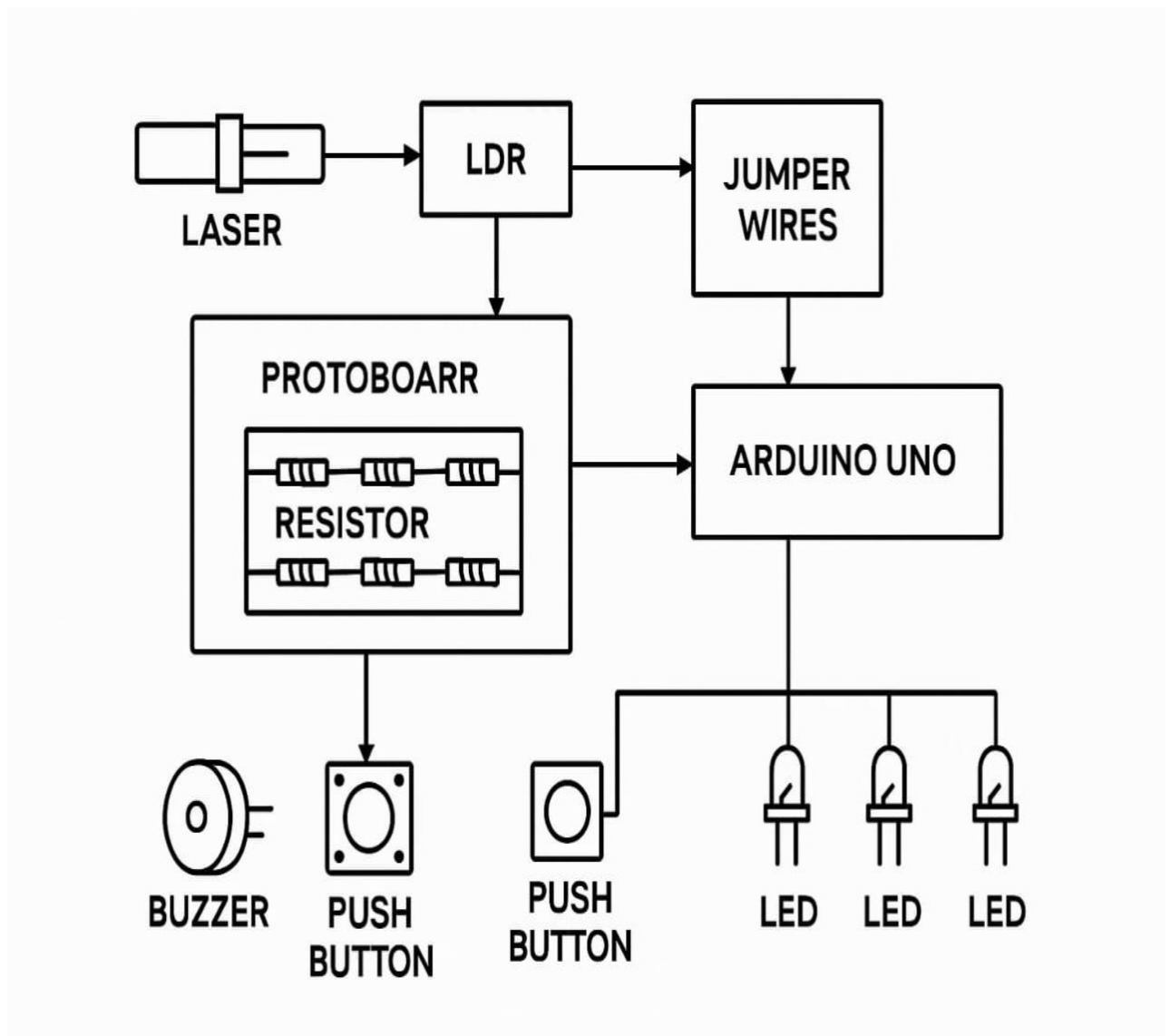
[4] Laser security system by Debarati Dutta. The proposed system contains LDR, IC555, transistor, buzzer which gives alert that the intruder has entered the monitored entrance through beep sound.

[5] Door lock security using Raspberry Pi and QR Code by Arigela Sai Kalyan, Balibineni Bharat Teja Raju, Mudraboina Venkatesh. In this project the study is aimed to develop a web-based security door using QR code system for the laboratory where at the same time the authorized person can perform access monitoring the student's attendance.

OBJECTIVE

The core objective of this project is to design a laser security system with laser and light dependent resistor. The laser based security alarm system is the same as its name suggest It will keep measuring the light coming from the laser light using LDR and will check its Initial value indicating obstacle between laser light and LDR the main component of this Project is LDR which detect dark.

BLOCK DIAGRAM:



METHODOLOGY

1. Laser Emitter

Purpose: Sends a focused laser beam to a fixed direction.

Working: This beam acts as an invisible barrier. When the beam is interrupted (e.g., by an intruder), the system detects the breach.

2. LDR (Light Dependent Resistor)

Purpose: Senses light from the laser.

Working: When the laser hits the LDR, it has low resistance. If the beam is blocked, its resistance increases, signaling the Arduino.

3. Arduino Uno

Purpose: Brain of the system.

Working: Takes input from the LDR. If it detects a change (laser beam interrupted), it activates the alarm (buzzer and LEDs).

4. Resistors (4x)

Purpose: Current limiting for LDR, LEDs, and button.

Placement:

1 for LDR voltage divider circuit.

1 for each LED to prevent overcurrent.

5. Soldering Pad / Breadboard

Purpose: Temporary platform to connect and test circuit components without soldering.

6. Jumper Wires

Purpose: Used to make electrical connections between components and the Arduino.

7. Push Button

Purpose: Manual reset or activation control.

Working: Can be used to reset the alarm or activate/deactivate the system.

8. LEDs (4x)

Purpose: Visual indicators.

Usage:

1 for system active status.

1 for intrusion alert.

1 for reset confirmation.

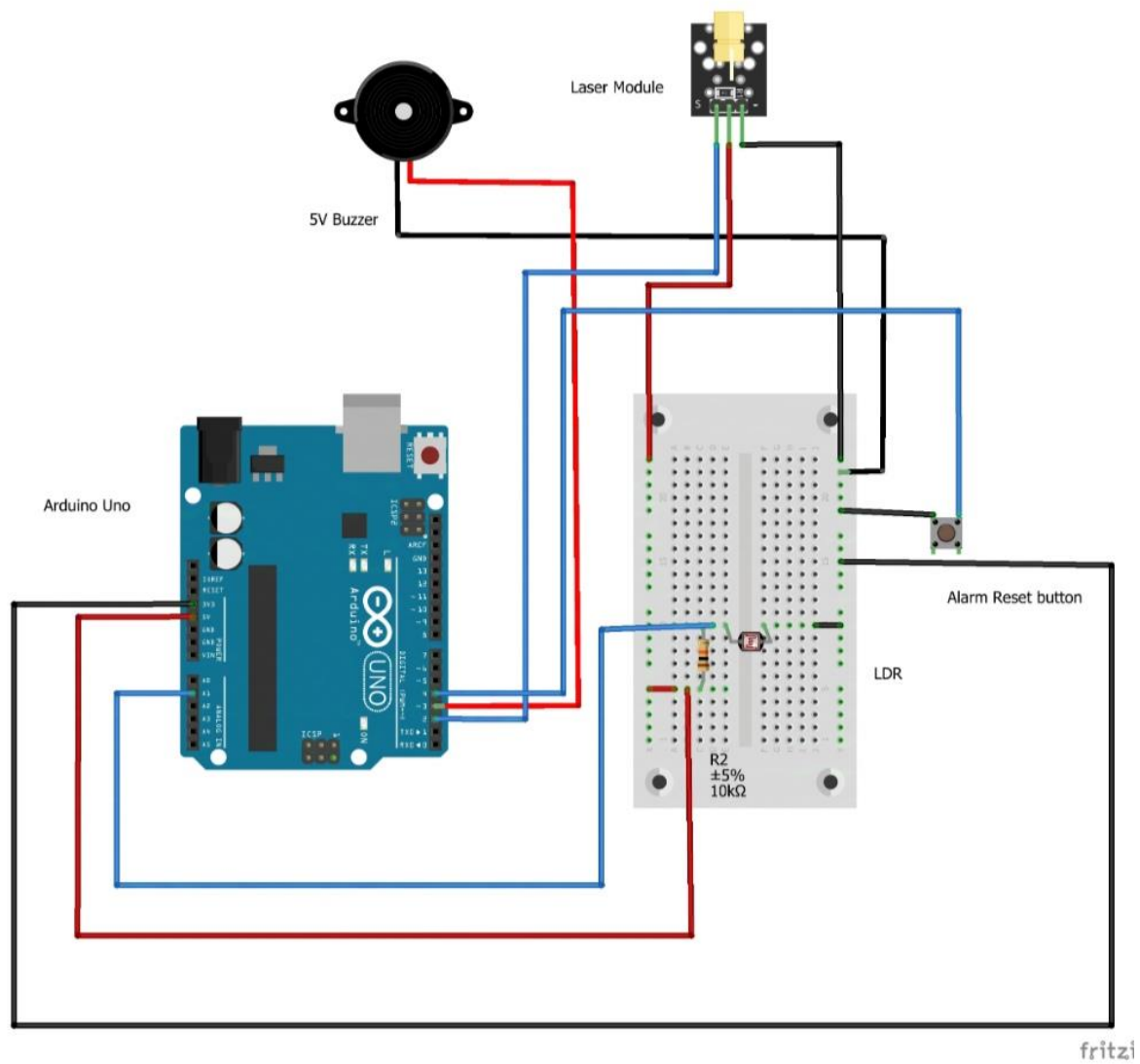
1 for laser alignment indicator.

9. Buzzer

Purpose: Audible alert when the laser beam is interrupted.

This setup simulates a basic intrusion detection system that can be used for doors, windows, or corridors. Would you like the Arduino code for this system too?

Circuit Diagram



COMPONENT DESCRIPTION:

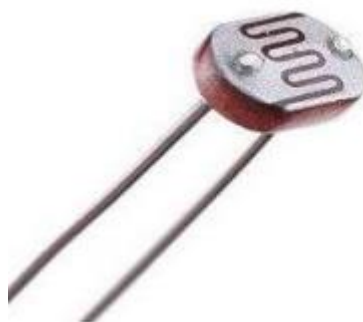
LASER:

A laser is a device that emits light through a process of optical amplification based on the stimulated emission of electromagnetic radiation. The term "laser" originated as an acronym for "light amplification by stimulated emission of radiation". A laser differs from other sources of light in that it emits light coherently. Spatial coherence allows a laser to be focused to a tight spot, enabling applications such as laser cutting and lithography. Spatial coherence also allows a laser beam to stay narrow over great distances (collimation), enabling applications such as laser pointers. Lasers can also have high temporal coherence, which allows them to emit light with a very narrow spectrum, i.e., they can emit a single colour of light. Temporal coherence can be used to produce pulses of light as short as a femtosecond.



LDR (Light Dependent Resistor):

A photo resistor or light-dependent resistor (LDR) or photocell is a light-controlled variable resistor. The resistance of a photo resistor decreases with increasing incident light intensity; in other words, it exhibits photoconductivity. A photo resistor can be applied in light sensitive detector circuits, and light- and dark activated switching circuits.



In the dark, a photo resistor can have a resistance as high as a few mega ohms (M ohms), while in the light, a photo resistor can have a resistance as low as a few hundred ohms.

BUZZER:

A buzzer or beeper is an audio signalling device, which may be mechanical, electromechanical, and piezoelectric. Typical uses of buzzers and beepers include alarm devices, timers and confirmation of user input such as a mouse click or keystroke. Early devices were based on an electromechanical system identical to an electric bell without the metal gong. Similarly, a relay may be connected to interrupt its own actuating current, causing the contacts to buzz. Often these units were anchored to a wall or ceiling to use it as a sounding board. The word "buzzer" comes from the rasping noise that electromechanical buzzers made.



LED (Light Emitting Diode):

A light-emitting diode (LED) is a two-lead semiconductor light source. Like an ordinary diode, the LED diode works when it is forward biased. In this case, the n-type semiconductor is heavily doped than the p-type forming the p-n junction. When it is forward biased, the potential barrier gets reduced and the electrons and holes combine at the depletion layer (or active layer), light or photons are emitted or radiated in all directions. A typical figure blow showing light emission due electron-hole pair combining on forward biasing.



RESISTOR:

A resistor is a passive two-terminal electrical component that implements electrical resistance as a circuit element. Resistors act to reduce current flow, and, at the same time, act to lower voltage levels within circuits. In electronic circuits resistors are used to limit current flow, to adjust signal levels, bias active elements, terminate transmission lines among other uses. High-power resistors that can dissipate many watts of electrical power as heat may be used as part of motor controls, in power distribution systems, or as test loads for generators. Fixed resistors have resistances that only change slightly with temperature, time or operating voltage.



CAPACITOR:

A capacitor (originally known as a condenser) is a passive two-terminal electrical component used to store energy electro statically in an electric field. The forms of practical capacitors vary widely, but all contain at least two electrical conductors (plates) separated by a dielectric (i.e. insulator). The conductors can be thin films, foils or sintered beads of metal or conductive electrolyte, etc. The non-conducting dielectric acts to increase the capacitor's charge capacity. A dielectric can be glass, ceramic, plastic films, air, vacuum, paper, mica, oxide layer etc.

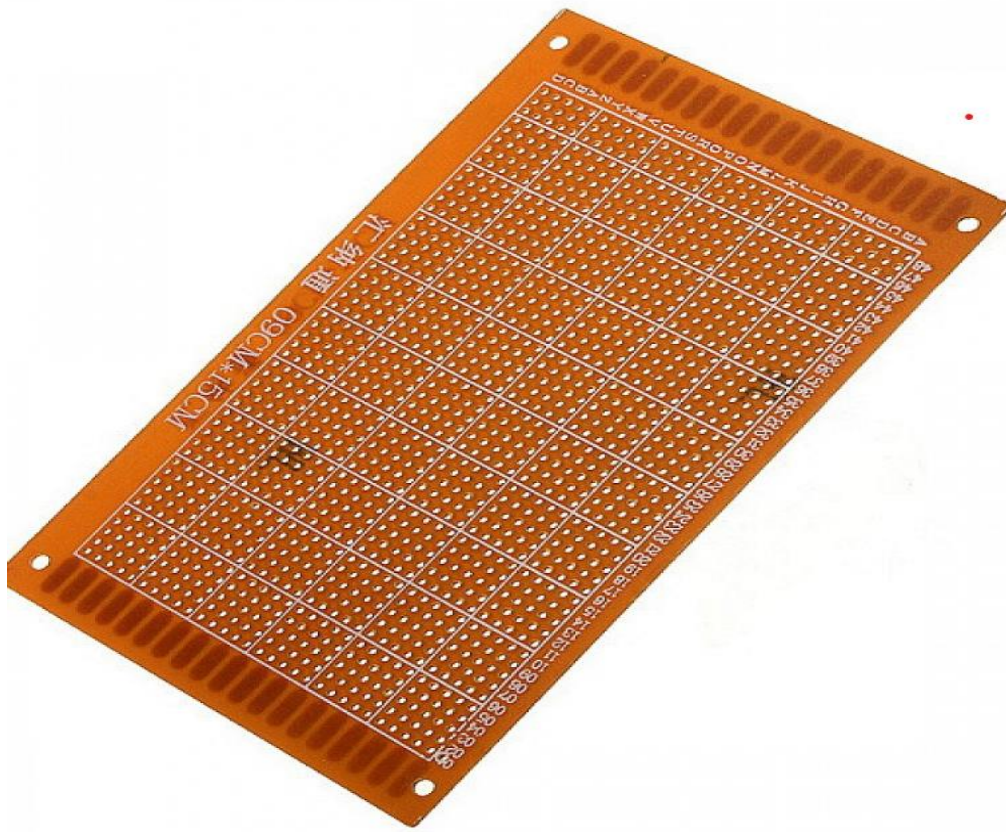


SOLDERING PAD AND CONNECTING WIRES:

Solder pads constitute one of the most vital elements on a printed circuit board facilitating reliable interconnections from components to copper traces during the soldering assembly process.

They represent termination points or localized metallized areas on the board providing wettable surfaces for anchoring and forming solder joints with component leads establishing both electrical continuity and mechanical anchorage.

This article provides a comprehensive perspective on various aspects related to solder pads including classifications, size calculations, spacing considerations, repair procedures and key design practices for optimal functionality.



Jumper Wires for solder less bread boarding can be obtained in ready to-use jump wire sets or can be manually manufactured. The latter can become tedious work for larger circuits. Ready-to-use jump wires come in different qualities, some even with tiny plugs attached to the wire ends. Jump wire material for ready-made should usually be solid copper, tin-plated wire - assuming no tiny plugs are to be attached to the wire ends. Shorter stripped wires might result in bad contact with the board's spring clips (insulation being caught in the springs).



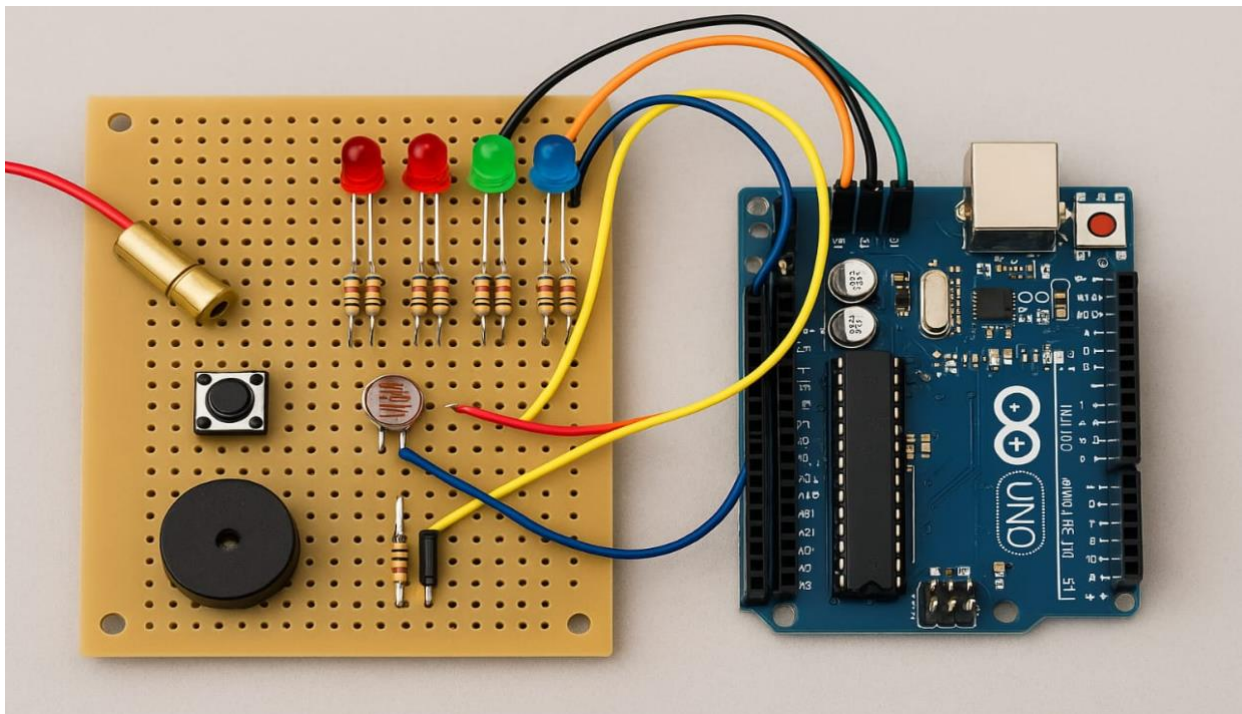
Arduino:

The Arduino Uno R3 is a microcontroller board based on a removable, dual-inline-package (DIP) ATmega328 AVR microcontroller. It has 20 digital input/output pins (of which 6 can be used as PWM outputs and 6 can be used as analog inputs). Programs can be loaded on to it from the easy-to-use Arduino computer program. The Arduino has an extensive support community, which makes it a very easy way to get started working with embedded electronics. The R3 is the third, and latest, revision of the Arduino Uno.



CIRCUIT DESIGN

The design of the laser security system circuit is very simple. Coming to the design of the circuit, First, the LDR and a 10 K Ω resistor are connected in series with the voltage divider and its output (common point) is connected to the pin 3 (non – inverting) of the Op- Amp IC LM358. For the inverting terminal (pin 2), connected the wiper of a 10 K Ω potentiometer (other two terminal of the POT are connected to VCC and GND). The 8th and 4th pin of Op-Amp IC LM358 are connected to the voltage divider and ground respectively. The output of the Op – Amp (Pin 1) is connected to the base of the transistor (BC547) through a resistor of 220 ohms. The trigger pin of 555 (Pin 2) is connected to the common point of 10 K Ω resistor and collector of transistor common point. The other end of resistor is given to voltage divider. The emitter of the transistor is given to the ground The reset pin (pin 4) of the 555 is connected to VCC through a 10 K Ω resistor and a push button is connected between Pin 4 of 555 and GND. The non-inverting input (5th pin) is connected to ground through 100nano farad capacitor. A buzzer is connected to pin 3 of ic555 IC. The other end of buzzer is given to the ground.



ADVANTAGES:

These are easy to install and work at both within as well as outside houses. These are very effective perimeter alarm systems around properties. In indoor systems can utilize the normal power outlets and jacks making them inconspicuous. At outside these can be easily be hidden behind the bushes or plants without causing any damage. They consume less power when compared to the laser system as the whole, which is expensive. These laser systems can be installed in homes either by self or by hiring a technical person. By technological innovations cost of the security systems has been cut to a large extent. So, making laser systems one among affordable security system options can be very safe. Lasers are strong in beam width and can be focused on the perfect target. By using laser security system one can be safe in the case of harmful effects to the body. As the beam width used in the laser security systems are not strong beam widths. The circuit, construction and setup for the Laser Security System are very simple. If used with a battery, the laser security system can work even when there is a power outage.

APPLICATIONS:

- Laser Security System can be used in safety lockers in our homes, where even if the locker's code is hacked, it acts as an additional layer of security.
- Apart from security systems, this laser based setup can also be used to check if pets or babies crossed a certain boundary.
- When the laser beam falling over the LDR is interrupted by the object in the field of laser net, hence the LDR develops an output voltage and the alarm rings showing the sign of any intruders.
- Laser security systems can be used in safety lockers to provide an additional layer of security.
- Laser security systems can be used in offices to alert you when someone is approaching.
- Laser security systems can be used to protect valuable items, such as paintings in a museum. They can create an invisible laser wall around the item, and alert you if someone tries to touch it.
- Laser security systems can be used to detect unauthorized entry into a home. They can be used in combination with other security systems.

CONCLUSION:

Laser security system provides us the security against any crime, theft in our day to day life and so people are installing them in order to stay safe, secure and sound. Various electronic security systems can be used at home and other important working places for security and safety purposes. It is a great opportunity and source of saving man power contributing no wastage of electricity. The "Laser Security System" is an important helping system. Using this system robbery, thefts & crime can be avoided to large extend. Avoiding thieves results in the safety of our financial assets and thereby this system provides us protection against all. The Laser & LDR system is highly sensitive with a great range of working. The system senses the light emitted by the Laser falling over the LDR connected with the circuit. Whenever the beam of light is interrupted by any means, it triggers the alarm or siren. This highly reactive approach has low computational requirement, therefore it is well suited to surveillance, industrial application and smart environments.

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